

V. General Website Requirements

The final application must contain at least **five (5) major pages or sections**.

The actual pages will depend on the selected business.

Suggested pages include:

- Home
- About Us
- Products/Services
- Contact Us
- Customer Transaction Page

Additional pages may include:

- Login
- Registration
- Product Catalog
- Shopping Cart
- Reservation
- Appointment
- Order Form
- Customer Dashboard
- Administrator Dashboard

Students are encouraged to develop pages that are meaningful to their chosen business rather than simply creating pages to satisfy the minimum number.

VI. Front-End Requirements

A. HTML

The application must demonstrate proper use of HTML.

Appropriate elements should be used for:

- Headings
- Paragraphs
- Images
- Hyperlinks
- Navigation
- Lists
- Tables
- Forms
- Buttons
- Sections
- Headers
- Footers

Pages should have a clear and logical structure.

B. CSS

The website must have a consistent visual design.

CSS should be used to implement:

- Color scheme
- Typography
- Spacing
- Navigation
- Buttons
- Forms
- Cards or content sections
- Page layouts
- Hover effects
- Responsive design

The website should remain usable on different screen sizes.

Students may use:

- Custom CSS
- Bootstrap
- Other approved CSS frameworks

However, students must understand and be able to explain the code used in their project.

VII. JavaScript Requirements

All projects must use JavaScript for client-side functionality regardless of the selected back-end technology.

Implement at least **three (3) meaningful JavaScript features**.

Examples:

- Form validation
- Dynamic price calculation
- Quantity controls
- Shopping cart calculation
- Search functionality
- Product filtering
- Image gallery
- Slideshow
- Modal windows
- Dynamic content
- Date validation
- Booking calculations
- Confirmation dialogs

The JavaScript features should contribute to the functionality or usability of the application.

Simple alert messages alone will not be considered sufficient implementation.

VIII. Database Requirements

The project must use a **MySQL database**.

The database must contain information relevant to the selected business.

Possible tables include:

- users
- customers

- products
- services
- orders
- order_items
- reservations
- appointments
- messages

The database must demonstrate appropriate use of:

- Primary keys
- Data types
- Field names
- Table names
- Relationships where appropriate

Students should avoid storing all application information in a single table when the data logically requires multiple related tables.

IX. Back-End Requirements

The application must have a working back end developed using either:

PHP

OR

Node.js with Express.js

The back end must communicate with the MySQL database.

The application must be able to receive information from the front end, process the information, communicate with MySQL, and return an appropriate result to the user.

Example:

Customer Form

↓

PHP / Node.js

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MySQL Database

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Record Successfully Saved

X. CRUD Requirements

The final application must demonstrate the four basic database operations.

CREATE

The application must allow users to add new records.

Examples:

- Add a product
- Place an order
- Make a reservation
- Book an appointment

READ

The application must retrieve and display records stored in the database.

Examples:

- View products
- View orders
- View reservations
- View customer information

UPDATE

The application must allow appropriate records to be modified.

Examples:

- Update customer information
- Edit a product
- Change order status
- Reschedule an appointment

DELETE

The application must allow appropriate records to be removed or cancelled.

Examples:

- Delete a product
- Cancel a reservation
- Remove an order
- Delete a customer record

XI. Input Validation

Forms must include appropriate validation.

Examples:

- Required fields cannot be empty.
- Email addresses must use an appropriate format.
- Quantities cannot be negative.
- Prices cannot be negative.
- Dates should be valid.
- Contact information should follow an appropriate format.

Validation may be implemented on both the client and server sides.

The application should display understandable error messages when invalid information is submitted.

XIV. Final Project Deliverables

Submit the following:

- Complete source code
- HTML files
- CSS files
- JavaScript files
- PHP files or Node.js files

MySQL database export (.sql)
Images/assets used by the application
Project documentation